

TokTickIT

E2E evidence and release integration

Student	GUNTEE DOUNGMANEE		
Student ID	67070501003		
Peer reviewer	POLWARIT	WATTHANAHEMMARAT	(MadMax168), ID 67070501067
Repository	ovenmakemeheat/toktickit		
Issue	#77 – Lab 3 E2E evidence and release integration		
Feature branch	feature/19-e2e-evidence-release		
Integrated target	lab3-staging, baseline 4cb1049		
Current final branch	main, eddafe67 (Lab 3 release PR #84 merged 2026-09-23)		
Stack	React, TypeScript, Vite, Bootstrap, Express, Prisma, PostgreSQL		

FINAL-MAIN CLOSEOUT – 2026-09-25. Release PR #84 merged Lab 3 into main at eddafe67, with MadMax168 as merger. The complete verification suite passed on staging commit d345420; its Git tree is identical to the release commit. A fresh local post-release rerun was partially blocked by unavailable Docker/PostgreSQL, as detailed in the evidence record; no unrun server or E2E check is reported as passing.

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1 Answer Part 1: Git Use with Engineering Workflow

Lab 3 used one feature branch per approved Issue and integrated feature Pull Requests into `lab3-staging`. The peer reviewer merged the feature PRs; the PR author did not merge their own work. The Issue #77 branch inherited the latest merged PR #82 rather than branching from an older feature branch. Release PR #84 then merged the integrated staging tree into `main`.

Issue / PR	Branch and merge state
#72 / #78	feature/14-sprint-specification; merged into lab3-staging by MadMax168.
#73 / #79	feature/15-authentication-and-requester; merged into lab3-staging by MadMax168.
#74 / #80	feature/16-it-staff-ticket-workflow; merged into lab3-staging by MadMax168.
#75 / #81	feature/17-administrator-user-management; merged into lab3-staging by MadMax168.
#76 / #82	feature/18-zen-green-visual-inspection; merged into lab3-staging by MadMax168.
#77 / #83	feature/19-e2e-evidence-release; merged into lab3-staging by MadMax168 on 2026-09-19.
Release / #84	lab3-staging to main; merged by MadMax168 on 2026-09-23 at eddafe67.

The completed two-stage sequence is: PR #83 integrated the evidence package into `lab3-staging`, then PR #84 released that tree to `main`. The final main commit and staging merge commit have identical Git trees. The reviewer record distinguishes the human merge evidence from formal GitHub approval state.

1.1 Repository, history, and review evidence

The engineering contract was first committed as `a4bd921` on 2026-09-10, before the Lab 3 implementation PRs. The final release merge is `eddafe67`; `d345420` is the identical verified staging tree.

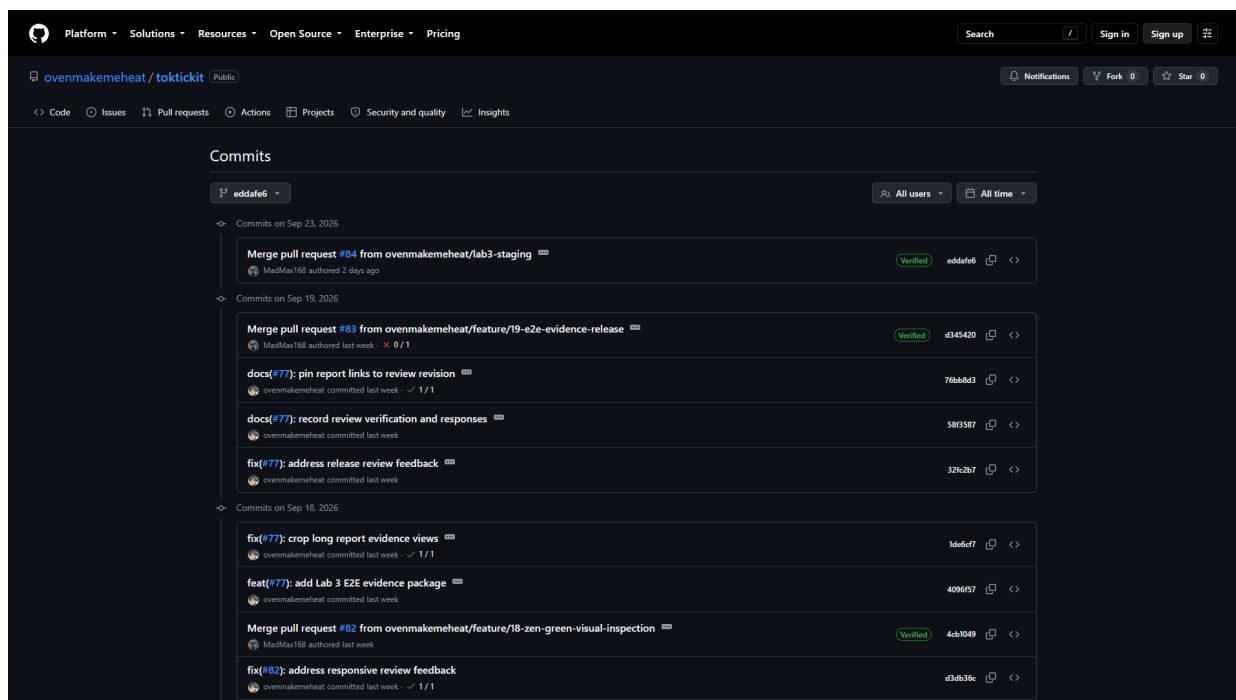


Figure 1: GitHub commit history at final-main release revision `eddafe67`, showing PRs #84 and #83 and the evidence-fix commits.

The linked [README](#) documents Bun/Docker setup, migration, seed, test database, and verification commands. The linked [.gitignore](#) excludes local environment files and secrets, local database/storage state, build output, test output, and LaTeX intermediates. Captures of the rendered final-main GitHub pages are saved in `docs/lab-03/report/evidence/`; the complete screenshots are retained there while content-focused views are used in this PDF.

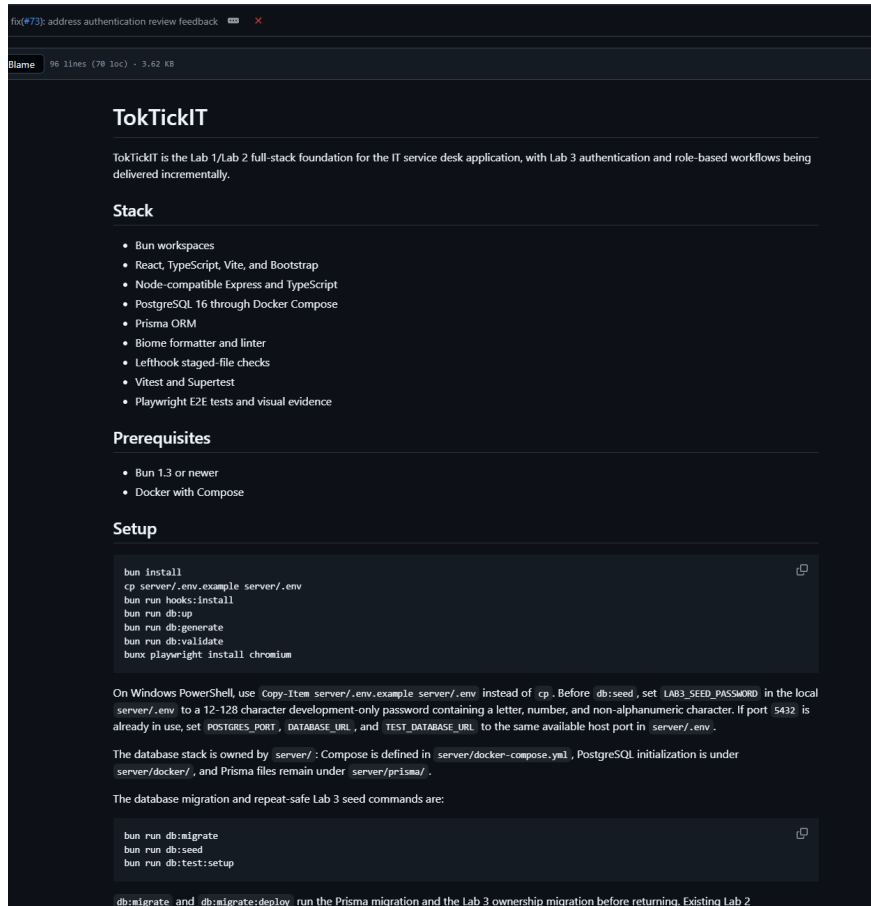


Figure 2: Rendered README on final main: project overview, stack, prerequisites, and setup (cropped; full capture retained).

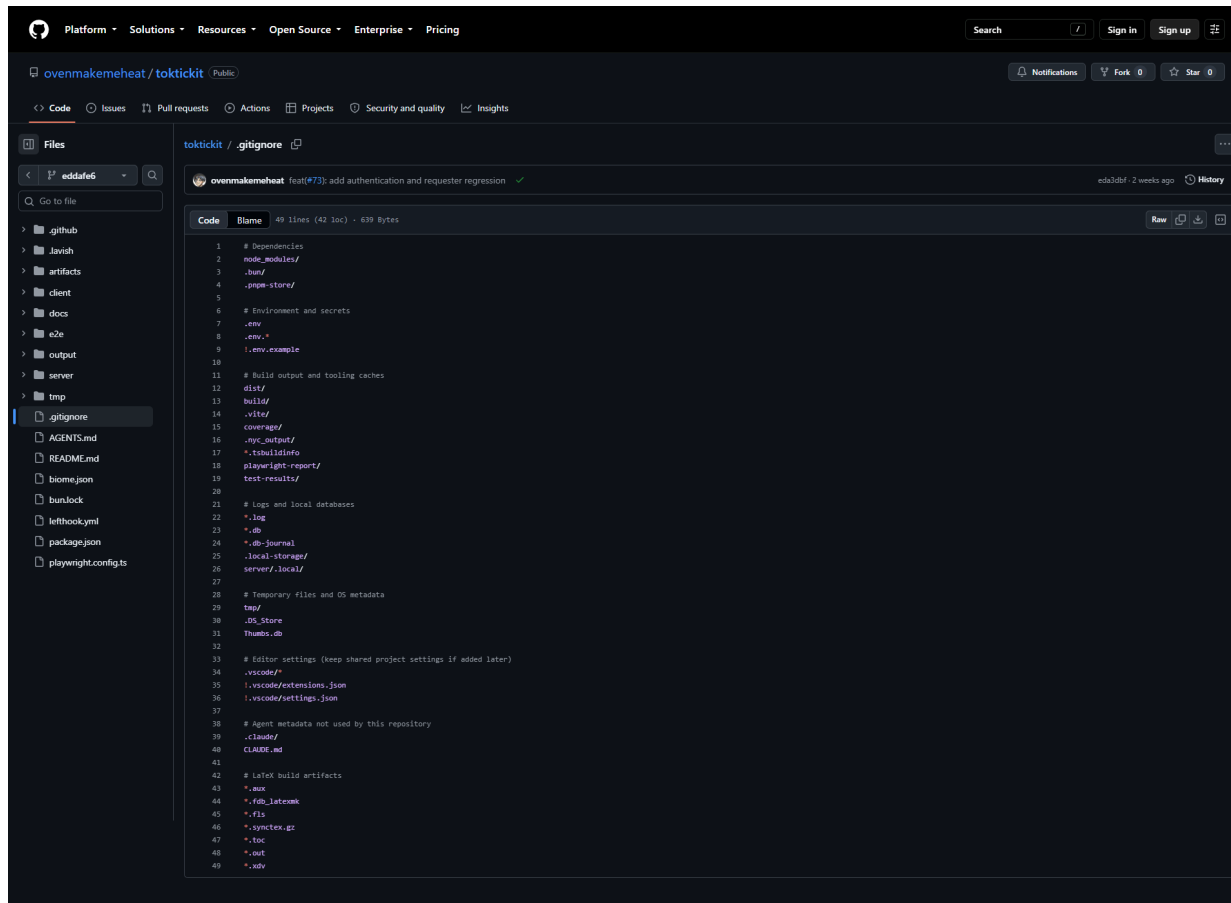


Figure 3: Rendered final-main .gitignore, including secrets, local state, test output, and LaTeX exclusions.

client/ (React app and Lab 3 UI/tests) server/ (Express, Prisma/migrations, Lab 3 API tests)
 e2e/lab-03/ (Playwright) docs/lab-03/ (specification, API/UI contracts, tests, evidence, report)
 artifacts/lab-03/screenshots/ (full-resolution role and viewport captures)

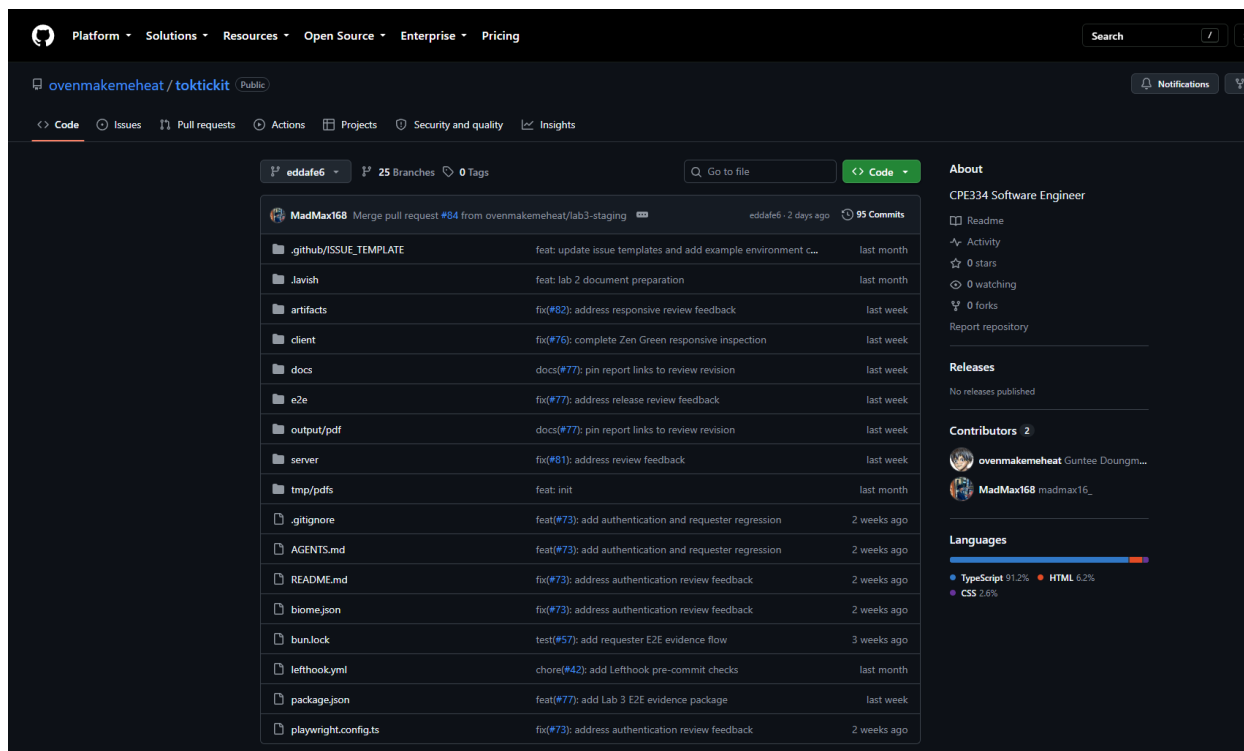


Figure 4: GitHub repository root at the Lab 3 release commit, showing the project folders, root files, and PR #84 merge.

The public [TokTickIT Lab 3 project board](#), [Kanban view \(Project #3\)](#) was captured read-only after release. It shows Issues #72–#77 in **Done**; no project fields were changed during this documentation work. The reviewer record names peer reviewer Polwarit Watthanhemmarat ([MadMax168](#)) and links PRs #78–#84; its review comments and responses are summarized below. The GitHub API returned no formal **APPROVED** decision for PRs #83 or #84, so no approval is invented here; PR #84’s human merger is recorded separately.

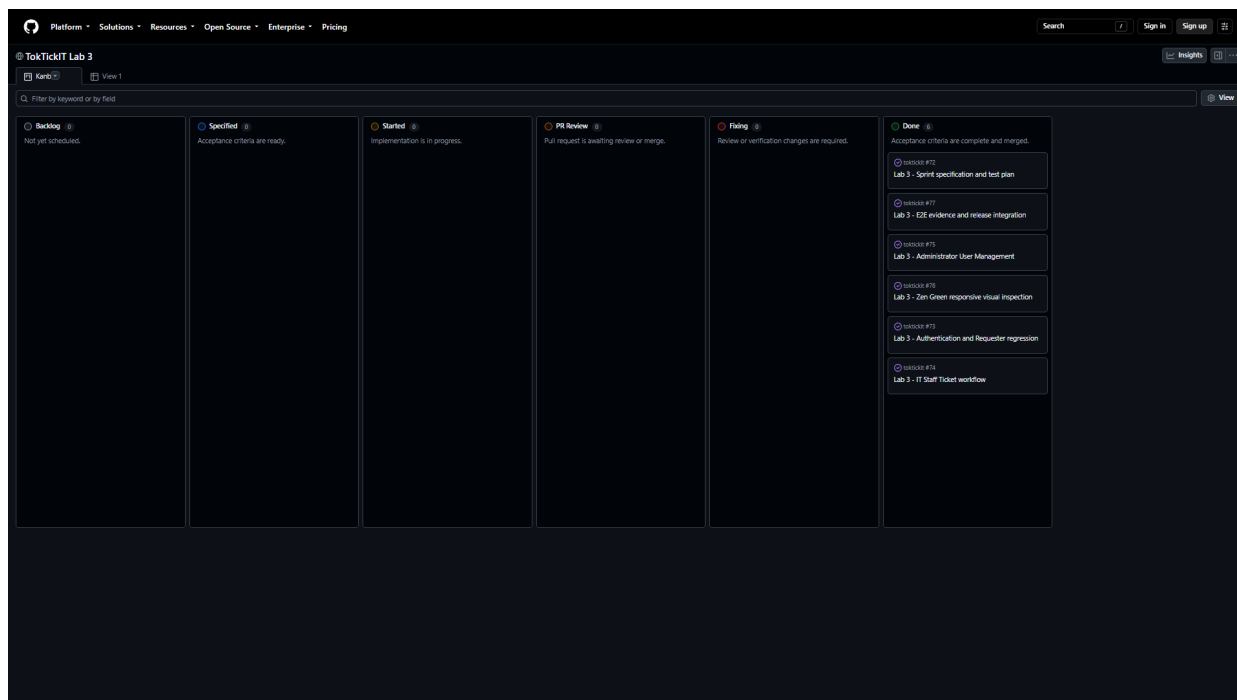


Figure 5: Public GitHub Kanban view, captured 2026-09-25; all six Lab 3 issue cards are in Done.

PR	Peer review topic	Author response / disposition
#78	Contract consistency, Administrator boundaries, migration handoff	Contracts and handoff were aligned; merged.
#79	Database-enforced Requester ownership, repeat-safe migration, Lab 2 regression	Ownership/migration flow and regression coverage were corrected; merged.
#80	Queue pagination, eligible owners, stale requests, concurrent status	Query/request handling and typed conflict feedback were corrected; merged.
#81	Mobile User Management, last-Administrator and assignment safety	Responsive cards and serialized account safeguards were added; merged.
#82	Responsive/accessibility assertions and missing Requester captures	Browser assertions and all nine Requester viewport captures were added; merged.
#83	Real seeded API evidence, immutable revision, report alignment	Unmocked seeded regression and review fixes were added; merged to staging.
#84	Release integration to main	Merged by MadMax168 ; no formal APPROVED review was returned by GitHub.

2 Answer Part 2: Specification DD

The contract is maintained in [docs/lab-03/specification.md](#) and is derived from the committed UTF-8 handout [UTF-8_Lab_3_sheet.pdf](#). The API, UI, and test contracts are linked from that specification. The contract’s first commit, [a4bd921](#) (2026-09-10), predates the feature implementation PRs; PR #78 records its review/follow-up.

The contract numbers FR-01–FR-32, BR-01–BR-29, and AC-01–AC-30. The requirements are summarized here; the linked specification contains the full rules, matrix, transition table, data model, migration plan, and Product Definition of Done.

Area	Contract IDs		Specified behavior / decision
Identity	FR-01-08; AC-01-08	BR-01-09;	Active-user login, opaque sessions, mandatory password change, CSRF, safe errors, and role navigation.
Migration / Re- quester IT Staff	FR-09-15; AC-09-14	BR-10-12;	Match Development Requesters by normalized email; preserve Ticket/Attachment ownership; derive new ownership from the session.
	FR-16-20; AC-15-21	BR-13-23;	Shared searchable Queue; eligible ownership; Requested/IT Priority separation; confirmed status matrix; public/private communication.
Administration	FR-21-26; AC-22-26	BR-24-26;	Minimal User list/search/filter/create/edit/reset with one role, duplicate-email, self-deactivation, and last-Administrator safeguards.
Quality / scope	FR-27-32; AC-27-30	BR-27-29;	Safe API errors, idempotent seed, responsive/accessibility states, test traceability, and explicit exclusions.

Role	Requester data	IT Staff workflow	Administrator access
Requester	Own Tickets/Attachments, Public Comments, resolution indication	None	None
IT Staff	None as Requester	Queue, ownership, status, priority, comments, notes	None
Administrator	None as Requester	No Queue/status/operational mutation	User Management; read-only Ticket Review and IT-Priority-only change

Selected acceptance criteria from AC-01–AC-30: AC-05 blocks normal screens until initial-password change succeeds; AC-09 preserves Ticket/Attachment ownership through migration; AC-20 accepts only confirmed status-matrix transitions; AC-21 limits Administrator Ticket Review to read-only data plus IT Priority; AC-26 protects self/last-Administrator status; and AC-28 requires usable, overflow-free desktop, tablet, and mobile screens.

Migration preserves Ticket and Attachment identity, maps Requester ownership by normalized email, hashes generated local-only initial passwords, and verifies counts/orphans before enforcing the new relationship. The Product Definition of Done ties these contracts to passing tests, review, responsive inspection, release evidence, and the single nine-part PDF; the formal-approval limitation is stated in Part 1 rather than silently marked complete.

Issue #77 does not add another role feature. It connects the already reviewed authentication/Requester, IT Staff, Administrator, and Zen Green increments to actual browser flows, evidence, review records, and release checks. The important boundaries are:

- Requester ownership comes from the authenticated session; the old Development Requester selector is absent.
- IT Staff may claim/reassign, change IT Priority, use the approved status matrix, post Public Comments, and add Internal Notes.
- Administrators may review existing Tickets and change only IT Priority from the review surface; they do not receive Queue or operational mutation controls.
- Browser fixtures prove visible workflows, while Supertest/PostgreSQL tests prove server authorization, migration, session, and account-safety rules.
- A reviewer comment, automation result, or generated artifact is not treated as human approval.

3 Answer Part 3: Test DD and Traceability

The executed matrix is recorded in [docs/lab-03/tests.md](#). The integrated release-tree snapshot passed on staging commit d345420; final main at edd4fe67 has the identical Git tree:

Boundary	Result
Client Vitest	15 files, 58 tests passed.
Server Vitest / Supertest / PostgreSQL	22 files, 143 tests passed.
TypeScript	Client and server type checks passed.
Playwright Lab 3	12 tests passed across authentication/Requester, Staff, Administrator, responsive, and release suites.
Build	Client and server builds passed.

The E2E files are listed below; together they cover valid and invalid credentials, inactive accounts, mandatory password changes, logout, Requester regression, Queue/detail operations, comments/notes, Administrator User Management, read-only Ticket Review, IT Priority, responsive controls, and direct role-route protection.

- `e2e/lab-03/authentication-and-requester.spec.ts`
- `e2e/lab-03/staff-ticket-flow.spec.ts`
- `e2e/lab-03/user-administration.spec.ts`

- `e2e/lab-03/responsive-and-accessibility.spec.ts`
- `e2e/lab-03/release-regression.spec.ts`

The test plan was written before implementation and maps every criterion to tests. The following ID ranges and actual paths render the plan's coverage summary; the linked `tests.md` contains the full per-test trace table.

- **UNIT-01-06 (Pass; AC-01-06, 09-10, 16, 20, 25):** password hashing, sessions/CSRF, migration, query parsing, and status transitions. Paths: `server/tests/lab-03/` (`auth.unit.test.ts`, `query.unit.test.ts`, `status-transition.unit.test.ts`, `migration.api.test.ts`).
- **API-01-16 (Pass; AC-01-27):** authentication, ownership, migration, direct authorization, Queue, communication, account safety, and safe errors. Path: `server/tests/lab-03/`; see the linked trace table for the individual files.
- **UI-01-09 (Pass; AC-02-08, 13-29):** labels/feedback, role shell, Requester regression, Staff/Admin screens, responsive and accessibility states. Paths: `client/tests/lab-03/` (`Login.test.tsx`, `ChangePassword.test.tsx`, `ApplicationShell.test.tsx`, `StaffTicketQueue.test.tsx`, `StaffTicketDetail.test.tsx`, `UserManagement.test.tsx`, `AdminTicketReview.test.tsx`).
- **E2E-01-06 (Pass; AC-01-08, 11-30):** authentication, Requester, Staff, Administrator, responsive/accessibility, and unmocked release regression. The five actual files are listed above under `e2e/lab-03/`.

The release-tree result is 58 client tests, 143 server tests, type checks and both builds passing, plus 12/12 Lab 3 E2E tests. The exact revision and the post-release local database limitation are recorded in [integration-verification.md](#).

4 Answer Part 4: AI Use with Reflection

The complete record is in [docs/lab-03/ai-use.md](#). OpenAI Codex coding agent ('gpt-5.6-luna', provider 'openai-codex', maximum reasoning) assisted with specification follow-up, E2E implementation, test execution, evidence writing, and report preparation.

Selected prompts used (7):

1. Read the Lab 3 repository instructions, issues, Pull Requests, and the PDF-derived contract before starting the next issue branch.
2. Verify that the next feature branch inherits the latest merged `lab3-staging` commit rather than an earlier feature branch.
3. Implement only Issue #77: browser E2E coverage for authentication, Requester regression, IT Staff operations, Administrator management, and release smoke checks.
4. Add actual final evidence records for test outputs, screenshots, review conversations, AI use, and the single Answer Part 1-9 PDF.
5. Check accessible roles/labels, direct role-route boundaries, queue/detail operations, public/private communication, and Administrator read-only boundaries in Playwright.
6. Run the focused E2E suites and then repository verification commands; record the exact branch, commit, prerequisites, and results without presenting a mocked browser test as proof of backend authorization.
7. After the release PR merged, reconcile the report and evidence records with the final `main` revision; distinguish the previously verified release tree from any post-release checks blocked by local infrastructure.

My Reflection. The specification-agent work translated the handout into numbered requirements, business rules, authorization boundaries, and testable acceptance criteria before implementation. The coding agent then stayed within that approved contract to implement integration tests and evidence. Keeping those roles distinct reduced the risk of deriving requirements from generated code. I reviewed the contract, diffs, screenshots, and test evidence and retain responsibility for human review responses and release decisions.

5 Answer Part 5: Authentication and Requester Regression

The authentication E2E flow starts from a protected route and verifies that an unauthenticated browser is sent to the sign-in screen. It exercises a valid login, generic invalid-credential feedback, a minimal inactive-account error, mandatory first-login password change, role navigation, logout, and the absence of the old Development Requester selector and Change Requester action.

The Requester regression keeps Create Ticket, My Tickets, Ticket Detail, and responsive overflow checks under the authenticated identity. Evidence covers valid/invalid credentials, inactive-account feedback, busy and safe-failure states, authenticated name/role navigation, mandatory password change, logout, and direct-route denial after logout. The browser tests are `e2e/lab-03/authentication-and-requester.spec.ts`, `client/tests/lab-03/Login.test.tsx`, and `client/tests/lab-03/ChangePassword.test.tsx`; Lab 2 Attachment and ownership API tests remain the evidence for server-side persistence.

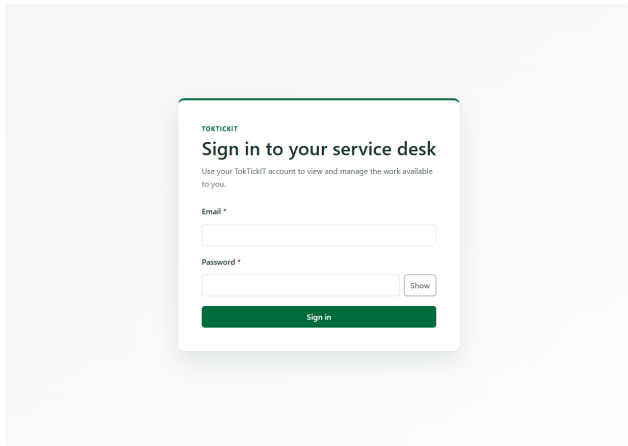


Figure 6: Login at desktop viewport.

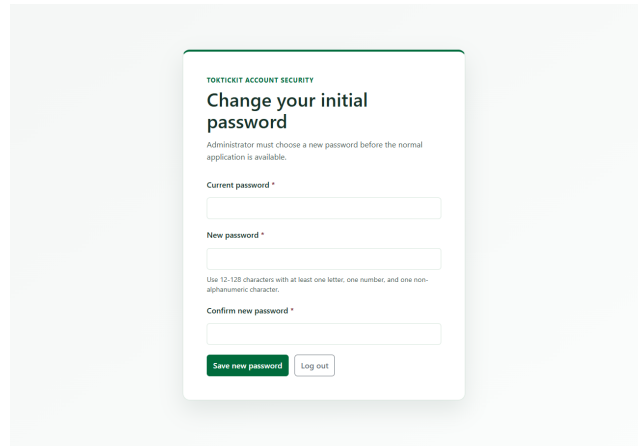


Figure 7: Mandatory initial-password change at desktop viewport.

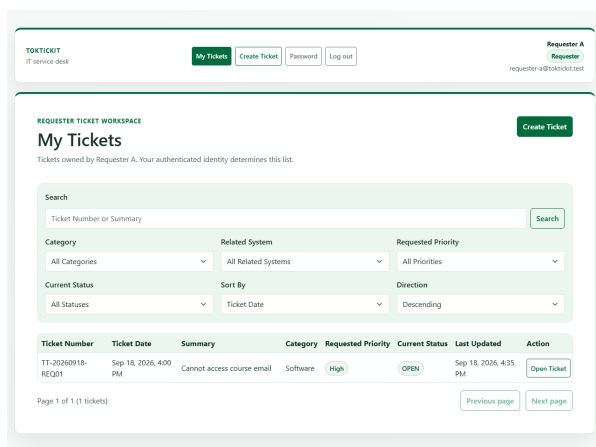


Figure 8: Requester My Tickets at desktop viewport.

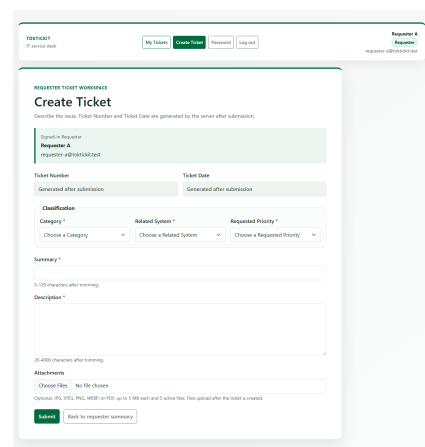


Figure 9: Authenticated Create Ticket at desktop viewport.

6 Answer Part 6: Working IT Staff Ticket Queue UI

The IT Staff Ticket Queue provides realistic seeded work with search, status, priority, owner, category, related-system, sort, and pagination controls. It shows Ticket Number, dates, summary, Requester, ownership, status, Requested Priority, and IT Priority without creating an unreadable mega-grid. The visual contract requires readable status/priority badges and a card representation at narrow widths. API tests cover deterministic queries, assigned/unassigned ownership, empty/no-results, invalid queries, and safe failures; client/E2E tests cover controls, badges, card/table behavior, and opening detail (`staff-queue.api.test.ts`, `client/tests/lab-03/StaffTicketQueue.test.tsx`, `e2e/lab-03/staff-ticket-flow.spec.ts`).

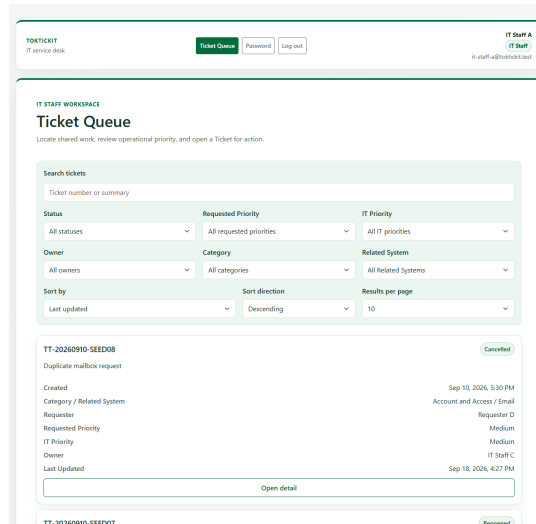


Figure 10: IT Staff Ticket Queue evidence (cropped top workflow view; full capture retained in the evidence directory).

7 Answer Part 7: Working IT Staff Ticket Detail UI

The IT Staff Ticket Detail keeps Ticket facts and existing Attachment metadata read-only while exposing only the approved operational controls: claim, reassign to an eligible owner, IT Priority, permitted status transitions, Public Comments, and Internal Notes. The screen distinguishes public and private communication, shows the Requester resolution indication, and provides validation, confirmation, conflict, and safe-failure feedback. API/client tests cover owner eligibility, CSRF, confirmation, concurrency, note visibility, and Attachment continuity (`server/tests/lab-03/staff-ticket-detail.api.test.ts`, `server/tests/lab-03/comments-notes.api.test.ts`). Direct role-forbidden API and destination checks are exercised by `e2e/lab-03/release-regression.spec.ts`; they do not rely on hidden buttons.

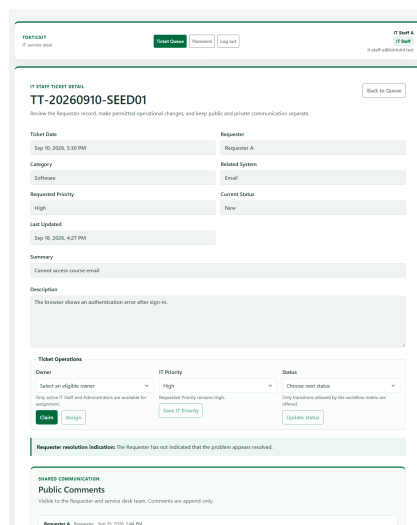


Figure 11: IT Staff Ticket Detail evidence (cropped top workflow view; full capture retained in the evidence directory).

8 Answer Part 8: Working Administrator User Management UI

The Administrator User Management screen remains intentionally minimalist. It shows Name, Email, Role, Status, and Edit; supports name/email search and one role filter; and provides create, basic edit, activation/deactivation, and initial-password reset. Browser and API tests cover duplicate email, invalid role/input, self-deactivation, last-active-Administrator protection, forced password change after reset, responsive presentation, safe failures, and forbidden access by non-Administrators. The User list, create form, and companion Ticket Review screenshots show the required fields and separate read-only Ticket surface with IT-Priority-only operation. The direct client/API coverage is in `client/tests/lab-03/UserManagement.test.tsx`, `client/tests/lab-03/AdminTicketReview.test.tsx`, and `server/tests/lab-03/users-admin.api.test.ts`.

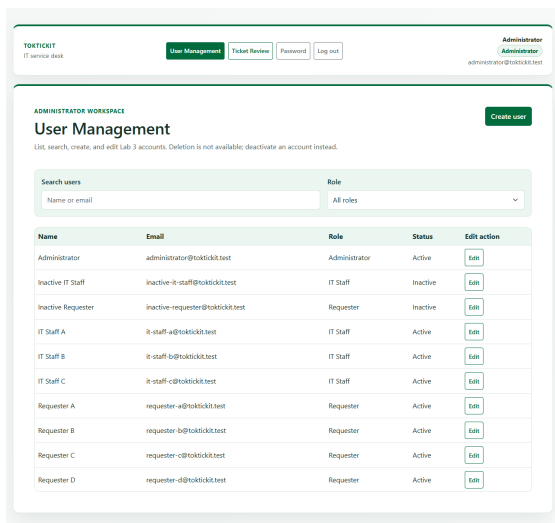


Figure 12: Administrator User Management evidence.

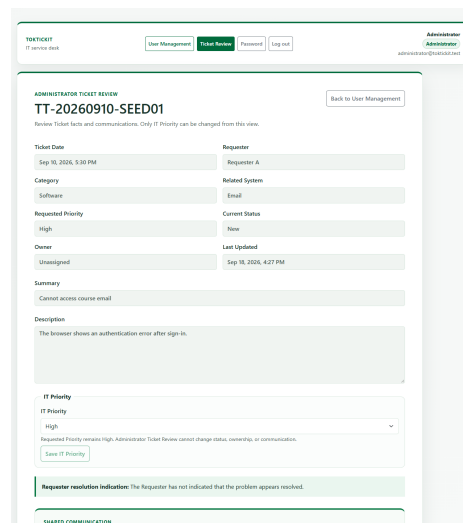


Figure 13: Administrator Ticket Review companion evidence (cropped top workflow view; full capture retained in the evidence directory).

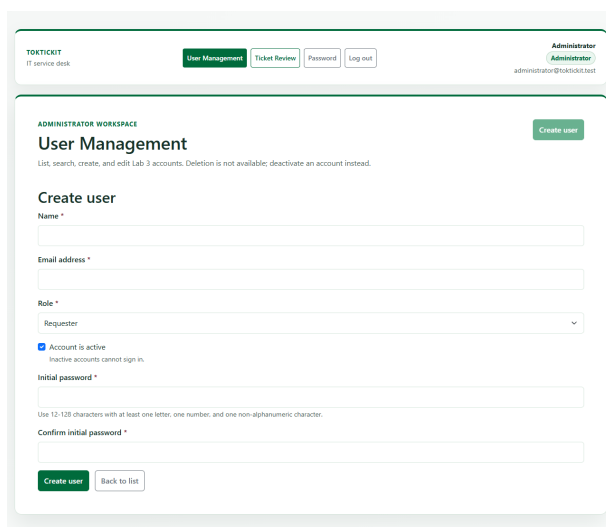


Figure 14: Administrator Create User form; full capture retained in the evidence directory.

8.1 Evidence, Review, and Release Integration

The required evidence inventory is linked in [integration-verification.md](#). It includes authentication, IT Staff Queue, Staff Ticket Detail, User Management, Requester, and desktop/tablet/mobile screenshot directories. The visual inspection checklist records the target viewports '1280x900', '820x900', and '390x844', no horizontal overflow, accessible names, focus, and the card/table representation.

[reviewer.md](#) records the student and peer reviewer identities, PRs #78–#84, review comments, author responses, and merge evidence. It keeps automated Codex and GitGuardian output separate from formal human approval. GitHub reports Issues #72–#77 closed at closeout; the public Lab 3 Project board capture in Part 1 shows all six in Done. The board was inspected read-only and not modified in this documentation update.

The release sequence is complete: PR #83 merged into `lab3-staging`, the integrated release tree passed verification, and PR #84 merged it into `main`. The post-release local verification limitation is recorded in the linked evidence file.

9 Answer Part 9: Zen Green, Responsive Evidence, and Limits

The visual contract is in [docs/lab-03/ui-spec.md](#). It applies Zen Green colors: primary #006B3C, secondary #0B7A46, and pale #EAF6EF. It also specifies readable system typography, a 4px spacing scale, grouped white surfaces, labels above fields, visible focus, and text-plus-color state cues. Loading, saving, validation, success, empty, no-results, forbidden, not-found, conflict, and safe-failure states are designed; responsive shells preserve role navigation and keep public/private communication distinct. The linked [completed visual checklist](#) records viewport

and overflow/accessibility inspection. The rendered UI-spec capture below is cropped to its overview; the complete screenshot is retained under `docs/lab-03/report/evidence/`.

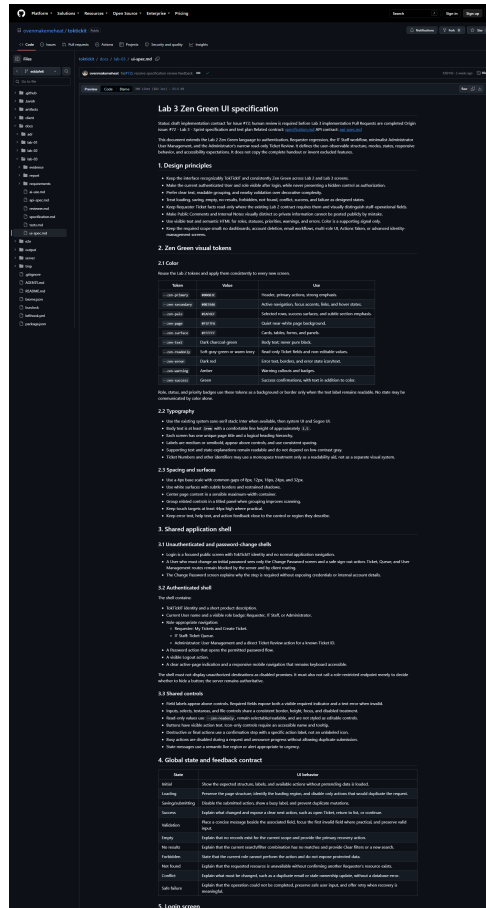


Figure 15: Rendered final-main `docs/lab-03/ui-spec.md` overview (cropped; full capture retained).

The desktop captures above are embedded. The following links open the original full-resolution desktop/tablet/mobile screenshots from the immutable release commit; long views are cropped only in the PDF, never in the evidence files.

Screen	Desktop	Tablet	Mobile
Login	view	view	view
Change Password	view	view	view
Requester My Tickets	view	view	view
Requester Create Ticket	view	view	view
Requester Ticket Detail	view	view	view
IT Staff Queue	view	view	view
IT Staff Ticket Detail	view	view	view
User Management	view	view	view
Create User	view	view	view
Administrator Ticket Review	view	view	view

Labels, live/status regions, named buttons, visible focus, required indicators, and non-color state cues are tested at the user-observable boundary.

Verification note. Lab 3 is released on `main` at `eddafe67`; this release has the same Git tree as the staging revision `d345420` that passed `bun run verify`, the 12-test Lab 3 Playwright suite, and the report build. A fresh local run on 2026-09-25 passed `lint/hooks/schema/type` checks, all 58 client tests, and both builds, but PostgreSQL-backed server tests and seeded E2E could not be rerun because Docker Engine was unavailable. The full local `bun run verify` attempt is documented as `environment-blocked`, not passed. The public Project #3 Kanban capture in Part 1 shows all six Lab 3 issue cards in `Done`; it was inspected read-only and no board fields were changed. No formal `APPROVED` review is claimed for PRs #83/#84.

No out-of-scope feature is claimed: there is no email delivery, social login, MFA, self-registration, User deletion, bulk User operation, SLA engine, notification service, dashboard, or production deployment work in this issue.